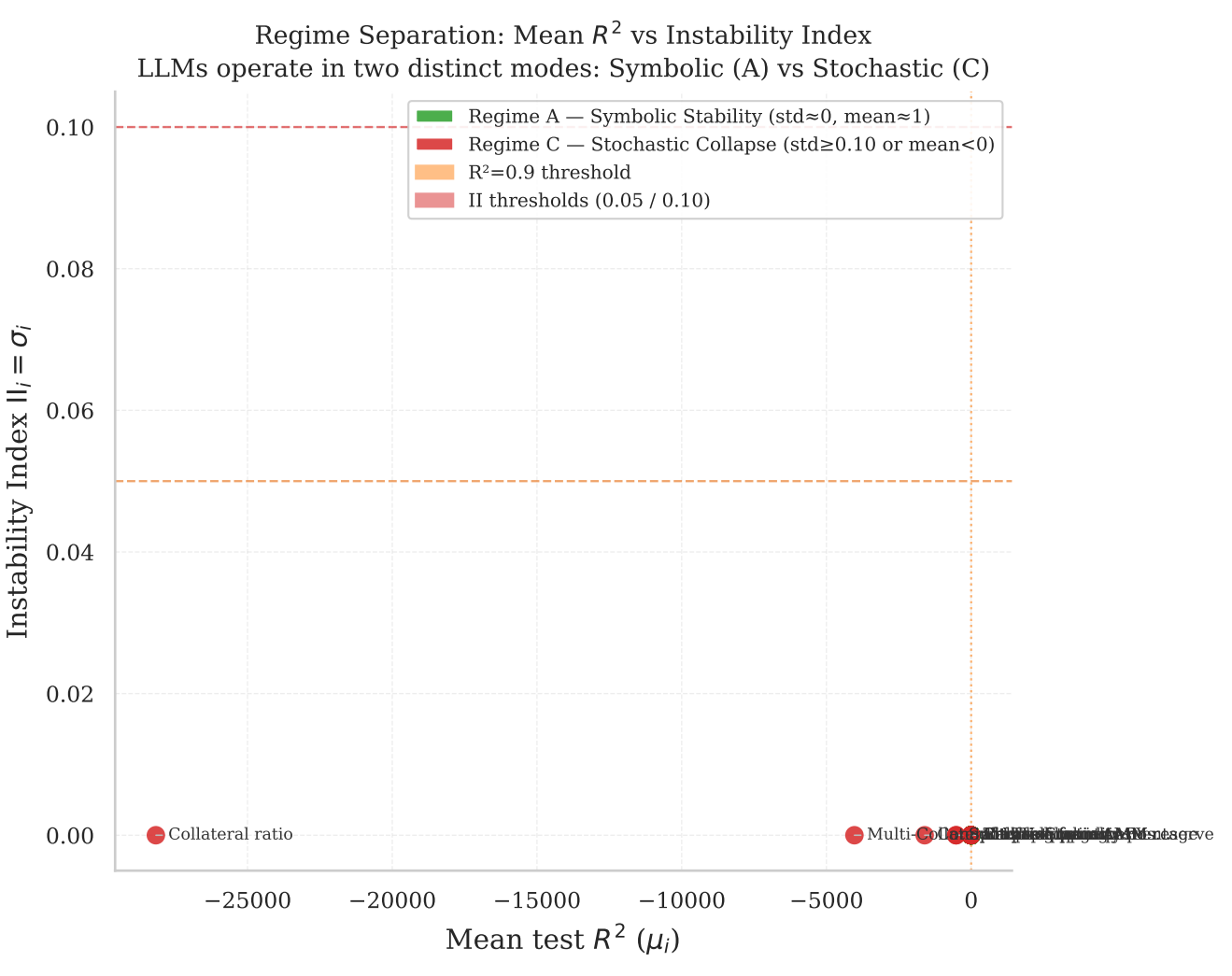
Regime Separation: Mean R^2 vs Instability Index
LLMs operate in two distinct modes: Symbolic (A) vs Stochastic (C)
Regime A — Symbolic Stability (std≈0, mean≈1)
Regime C — Stochastic Collapse (std≥0.10 or mean<0)
R²=0.9 threshold
II thresholds (0.05 / 0.10)
Instability Index $II_i = \sigma_i$
0.10
0.08
0.06
0.04
0.02
0.00
Collateral ratio
−25000
−20000
−15000
−10000
−5000
0
Mean test R^2 (μ_i)